

Distribution of ABO and Rh Blood Group Systems Among Blood Donors in a Tertiary Hospital, North-Western Nigeria

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ABSTRACT

Out of the 45 blood group systems currently recognised by the International Society of Blood Transfusion, the ABO and Rh are still considered the most important in clinical practice. Knowledge of these blood group systems has improved our understanding of blood transfusion, transplantation and neonatal haemolytic disease. The distributions of ABO and Rh blood group systems have been shown to vary among populations depicting the effects of genetics and environment. This study was aimed to determine the distribution of ABO and Rh blood groups among blood donors in a tertiary health facility in North-western Nigeria. This was a cross-sectional study involving blood donors who had their ABO and Rh blood group systems determined via Ag-Ab agglutination technique using the tile method. A total of 1,022 blood donors were recruited with age range of 18 to 67 years and predominantly male 910 (89.0%). The distributions of the blood group systems were: ABO: O 595 (56.2%), B 205 (20.1%), A 192 (18.8%) and AB 30 (2.9%); and Rh: Rh D positive 916 (89.6%) and Rh negative 106 (10.4%). Twenty (2.0%) of the female subjects were found to be Rh negative. Our study found the distribution among blood donors of ABO blood group system in descending order as O, B, A, and AB; while for Rh blood group, majority were Rh positive. These findings could be utilized in organization of blood donor drive, improving safety of blood transfusion services and mapping out screening services for neonatal haemolytic disease.

Keywords: ABO, Rh, Blood group systems, Blood donors, Nigeria

INTRODUCTION

A blood group system is defined by International Society of Blood Transfusion (ISBT) as one or more antigens governed by a single gene or complex of two or more closely linked homologous genes.¹ In ISBT currently 45 and 360 blood group systems and RBC antigens are recognized respectively.² The ABO and Rh are still considered as the most important blood group systems in clinical practice largely due to their applications in blood transfusion medicine, transplantation and neonatal haemolytic disease.^{3,4}

The ABO and Rh blood systems are the most critical in transfusion medicine, and their distribution varies significantly across different regions.^{5,6} A study conducted in Ghana revealed that 57.4% of the participants had blood group O positive, while a similar study by Oyiola et al in Lagos reported a

prevalence of 52.9% of same blood group.^{7,8}

The ABO and Rh blood systems plays a vital role in blood transfusion practice, ensuring compatibility between donor blood and the recipient's blood. Both ABO and Rh blood group systems guide blood transfusion decisions, ensuring the most compatible is transfused to recipients.^{9,10,11} The combination of ABO and Rh blood systems helps prevent transfusion related complications, such as haemolytic disease of the newborn (HDN) and transfusion related acute lung injury (TRALI).¹²

Transfusing ABO or Rh incompatible blood can lead to severe and potentially life-threatening consequences which may be immediate or delayed. The immediate effects of incompatible group transfusion include acute haemolysis, shock, hypotension and renal failure, any of which can cause death if not identified and treated promptly.^{10,13,14,15}

The delayed effect of administering incompatible blood may occur in approximately three months after transfusion in patients who have received blood transfusion in the past and have experienced a second re-exposure.¹² The clinical presentation complication can

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include haemolysis, anaemia and jaundice.^{12,16} The variability of the ABO and Rh antigen can trigger massive activation of the immune system and clotting system resulting in shock, kidney failure, circulatory collapse and potentially death in some individuals.¹⁷

Variations in the distribution of ABO and Rh blood group systems among different populations have been reported in the literature,^{18,19} thus our study aimed to determine the distribution of ABO and Rh blood group systems in our local tertiary health centre in North-western Nigeria.

MATERIALS AND METHODS

Study design, study area and study population

This was a prospective descriptive study which involved consecutive recruitment of consenting blood donors at the bleeding bay of Ahmadu Bello University Teaching Hospital (ABUTH) Zaria, Kaduna State North-western Nigeria over a period of four months (1st April 2017 to 31st July 2017). ABUTH Zaria serves as a tertiary health care facility that provides care for patients within the North-western region of Nigeria and even beyond.

Determination of blood group systems

Four milliliters (4mls) of venous blood were aseptically collected from all study subjects; and of this, three milliliters (3mls) were delivered into tripotassium ethylenediamine tetra acetic acid (K3-EDTA) bottle while the remaining one milliliter (1ml) was dispersed into a plain sample bottle.²⁰ Both ABO and Rh blood groupings were determined via Ag-Ab agglutination technique using the tile method.¹⁸ This entailed mixing of a drop of the subject's whole blood with two drops of commercially obtained anti-A, anti-B, anti-AB and anti-D antisera on a plastic tile (forward grouping). While the antisera manufacturer's instructions were adhered to, a glass rod was used to thoroughly mix the blood and the antisera while the tile was rocked gently. Agglutinations were looked for both

macroscopically using hand-held glass lenses and microscopically after 2-5 minutes. In unclear cases of agglutination, reverse grouping using the donor serum and known groups A and B cells was performed for ABO blood group system determination; while incubation of the mixture in a test tube at 37°C for 15 minutes was employed for Rh blood group determination.

Data collection, data analysis and data representation

A proforma was used to collect the subjects' age and gender while the ABO and Rh blood grouping outcomes were also recorded. Data generated were eventually entered into Statistical Package for the Social Sciences (SPSS) version 20 for analysis. Results of the data analysis were presented in form of proportions, frequencies and percentages which were subsequently tabulated.

Ethical consideration

All subjects gave informed consent to partake in the study while ethical approval for the study was obtained from the Health Research Ethical Committee (HREC) of ABUTH Zaria, Kaduna State, Nigeria.

RESULTS

A total of 1,022 donors were recruited for the study and had an age range of 18 to 67 years with gender distribution of male 910 (89.0%) and female 112 (11.0%). For ABO blood group system, blood group O had the highest 595 (58.2%) while AB blood group had the least 30 (2.9%). With regards to Rh blood group system, Rh positivity was recorded in 916 (89.6%) of the study subjects. The detailed ABO and Rh blood group systems distribution is as depicted in Figure 1. A total of 20 (2.0%) of the female subjects were found to be Rh negative with regards to the distribution of Rh blood group system. Table 1 further highlight on the distribution of ABO and rhesus blood group systems according to the gender of the study subjects.

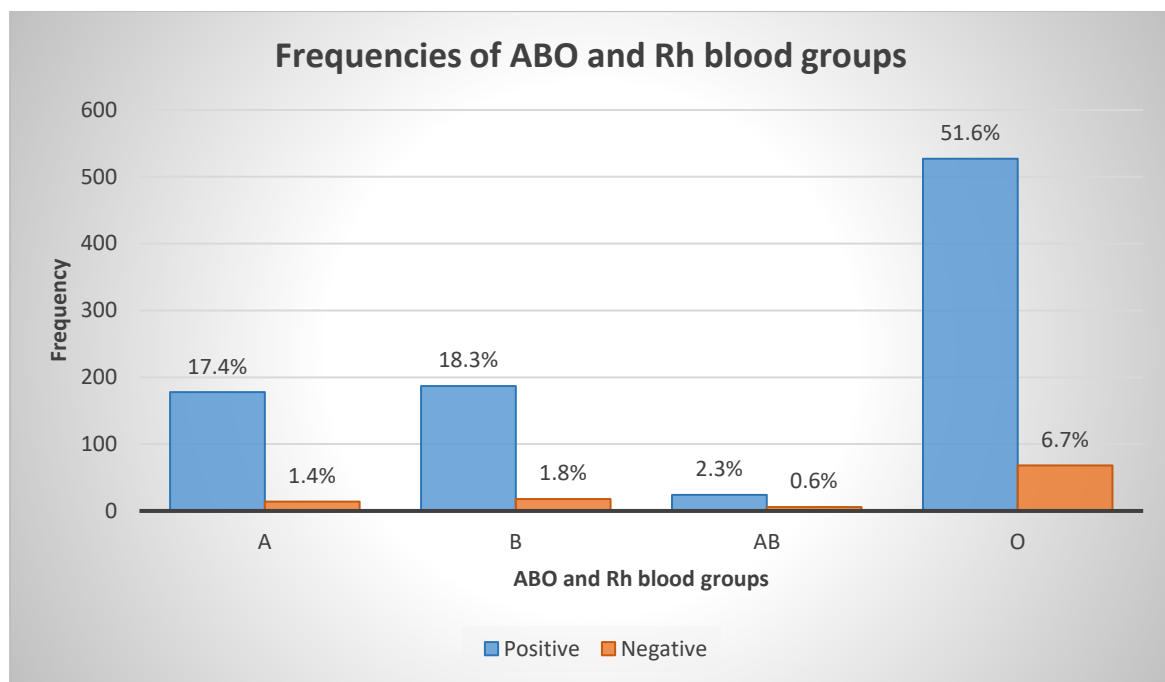


Figure 1: Frequencies of ABO and Rh blood group systems among the blood donors

Table 1: Distribution of ABO and Rh Blood group systems based on gender

ABO blood group	Rhesus blood group system				Total n(%)
	Positive		Negative		
system	Male	Female	Male	Female	
A	158	20	10	4	192 (18.8)
B	165	22	16	2	205 (20.1)
AB	22	2	2	4	30 (2.9)
O	479	48	58	10	595 (58.2)
Total n(%)	824	92	86	20	1022 (100)

DISCUSSION

A blood group is an individual's combination of red blood cell (RBC) surface antigens. It is the presence and absence due to inherited variation of RBC surface antigen that defines the blood group of an individual.

A blood group system is defined by International Society of Blood Transfusion (ISBT) as one or more antigens governed by a single gene or complex of two or more closely linked homologous genes.¹ According to ISBT, currently 45 and 360

blood group systems and RBC antigens are recognized respectively.²

The ABO and Rh are still considered as the most important blood group systems in clinical practice largely due to their applications in blood transfusion medicine, transplantation and neonatal haemolytic disease.^{3,4} Some researchers have also reported associations between certain human diseases with particular blood groups.^{16,17,18} Variations in the distribution of ABO and Rh blood group systems among different populations have also been reported in the literature,^{18,19} probably reflecting the interplay between genetic and environmental factors in their determination.

Our findings of the occurrence of ABO blood group system in the pattern of O>B>A>AB is similar to that of an earlier study by Musa et al in Sokoto; a town also located in the North-western Nigeria and of similar sociodemographic population with our study location.¹⁸ The separate studies of Faduyile et al in Lagos South-western Nigeria and that of Enosolease et al in Benin Southern Nigeria concurred with ours with regards to blood groups O and AB having the highest and least distribution respectively, but were at variance with ours in reporting blood group A as the second most common blood group among their study subjects.^{24,25} The predominance of blood group O in the Nigerian population had earlier been reported by a review conducted by Anifowoshe et al.²⁶ Studies from African populations such as that of Mohamud et al., in Somalia and Enawgaw et al., in Ethiopia are similar to ours with regards to blood group O with the highest frequency and AB the least; albeit blood group A was reported as the next most common after blood group O.^{27,28}

With regards to the distribution of Rh blood group system, our finding of majority of our study subjects been Rh positive is in agreement with quite a number of earlier studies in Nigeria and beyond.^{18,24-28} So far, all the previous studies we cited concurred with ours in the predominance of the Rh positivity in the studied population. While we recorded 89.6% Rh positivity, the studies of Mohamud et al., Musa et al., Anifowoshe et

al., and Enawgaw et al., had an overwhelming 97.0% 96.4%, 94.9% and 92.5% distribution respectively.^{18,24-28}

When the distributions of both ABO and Rh blood group systems are considered together, our recorded pattern of distribution turned out to be in the following sequence: O Rh+ve>B Rh+ve>A Rh+ve>O Rh-ve>AB Rh+ve>B Rh-ve>A Rh-ve>AB Rh-ve. Our study observed, with regards to gender, similar pattern of distribution for ABO and Rh positivity as O Rh+ve>B Rh+ve>A Rh+ve>AB Rh+ve. However, the ABO and Rh negativity distribution pattern for male was O Rh-ve>B Rh-ve>A Rh-ve>AB Rh-ve which differed from that of the female that was O Rh-ve>A Rh-ve/AB Rh-ve, B Rh-ve>. In this regard, while some studies could not establish any difference in the distribution of ABO and Rh blood group systems with regards to gender,^{18,29} others did observe some gender specific variation.³⁰

CONCLUSION

Our study has provided the ABO and Rh blood group system for a tertiary hospital in North-western Nigeria as follows: O Rh+ve>B Rh+ve>A Rh+ve>O Rh-ve>AB Rh+ve>B Rh-ve>A Rh-ve>AB Rh-ve; findings that varied from observations by other workers probably reflecting effect of genetics and environment. Our findings could be useful in the organization of blood donor drive, improving safety of blood transfusion services, mapping out screening services for neonatal haemolytic disease and understanding the possible relationship between these blood group systems and the prevalence of some human diseases.

Limitation

Only phenotypic determination of the ABO and Rh blood group systems was conducted and not the genotypic typing. Furthermore, the tile method employed in this study, which is more readily available in our setting, may not be as efficient as the more standard microplate agglutination method.

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Conflict of interest

There are no conflicts of interest.

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